Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 0.777858
<D> / site : 0.129643

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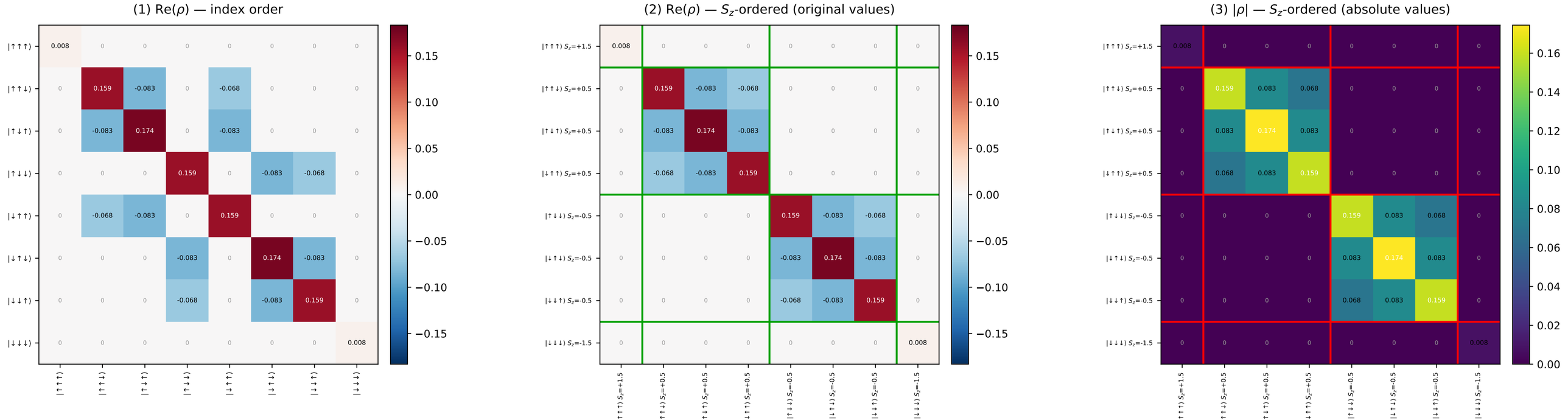
Body order : 3
Clusters   : 20

Date: 2026-07-29 11:32

Summary — 3-Body RDMs (Benzene, CAS(6,6))

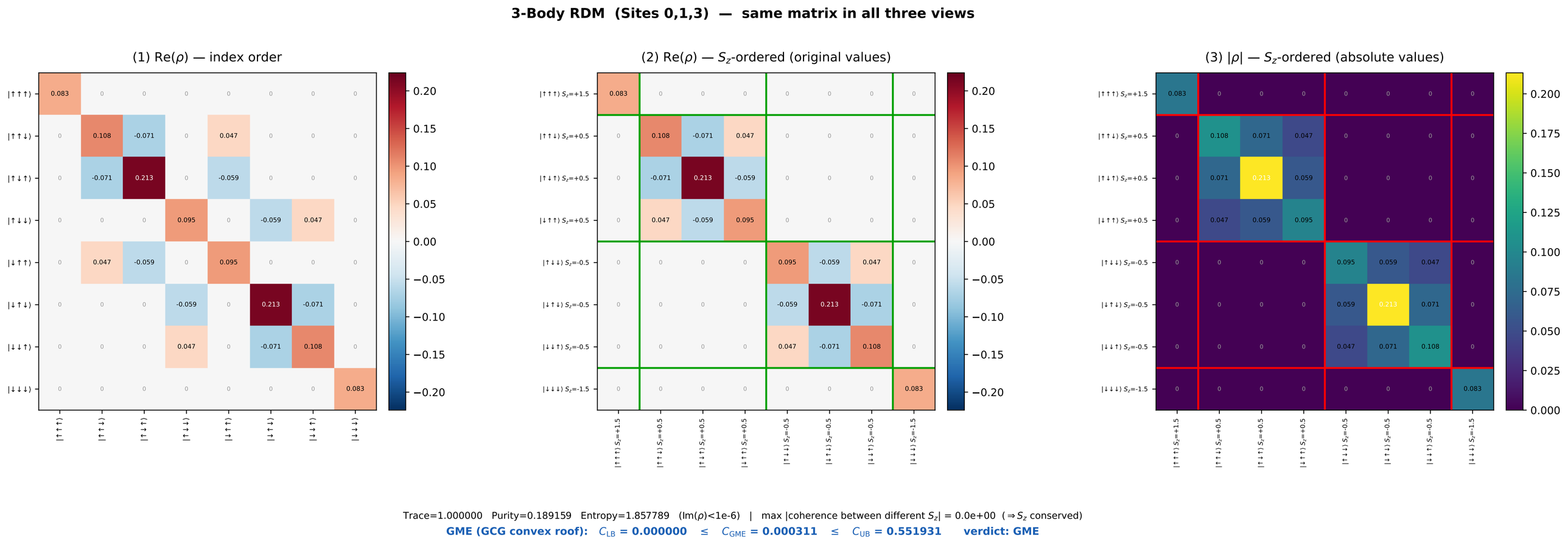
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.235436	1.526298	—	0.000000	0.000458	0.399163	MIXED
0,1,3	0.189159	1.857789	—	0.000000	0.000311	0.551931	STRONGLY MIXED
0,1,4	0.189159	1.857789	—	0.000000	0.000023	0.551931	STRONGLY MIXED
0,1,5	0.235436	1.526298	—	0.000000	0.000022	0.399163	MIXED
0,2,3	0.189159	1.857789	—	0.000000	0.000012	0.550327	STRONGLY MIXED
0,2,4	0.173594	1.869971	—	0.000000	0.000256	0.391258	STRONGLY MIXED
0,2,5	0.189159	1.857789	—	0.000000	0.000278	0.551931	STRONGLY MIXED
0,3,4	0.189159	1.857789	—	0.000000	0.000006	0.551931	STRONGLY MIXED
0,3,5	0.189159	1.857789	—	0.000000	0.000074	0.551931	STRONGLY MIXED
0,4,5	0.235436	1.526298	—	0.000000	0.000553	0.399153	MIXED
1,2,3	0.235436	1.526298	—	0.000000	0.001594	0.399163	MIXED
1,2,4	0.189159	1.857789	—	0.000000	0.000000	0.550740	STRONGLY MIXED
1,2,5	0.189159	1.857789	—	0.000000	0.000007	0.551931	STRONGLY MIXED
1,3,4	0.189159	1.857789	—	0.000000	0.000002	0.550809	STRONGLY MIXED
1,3,5	0.173594	1.869971	—	0.000000	0.000256	0.391258	STRONGLY MIXED
1,4,5	0.189159	1.857789	—	0.000000	0.000006	0.551931	STRONGLY MIXED
2,3,4	0.235436	1.526298	—	0.000000	0.001176	0.399160	MIXED
2,3,5	0.189159	1.857789	—	0.000000	0.000008	0.551093	STRONGLY MIXED
2,4,5	0.189159	1.857789	—	0.000000	0.000013	0.547662	STRONGLY MIXED
3,4,5	0.235436	1.526298	—	0.000000	0.000156	0.399059	MIXED

3-Body RDM (Sites 0,1,2) — same matrix in all three views

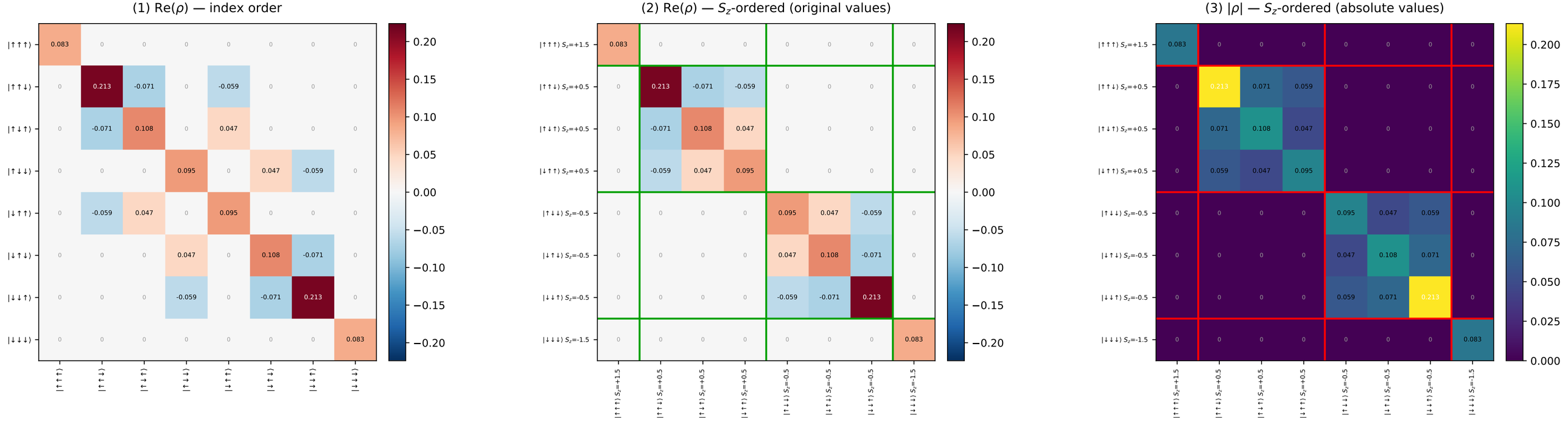


Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000458 ≤ C_{UB} = 0.399163 verdict: GME


3-Body RDM (Sites 0,1,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000311 ≤ C_{UB} = 0.551931 verdict: GME

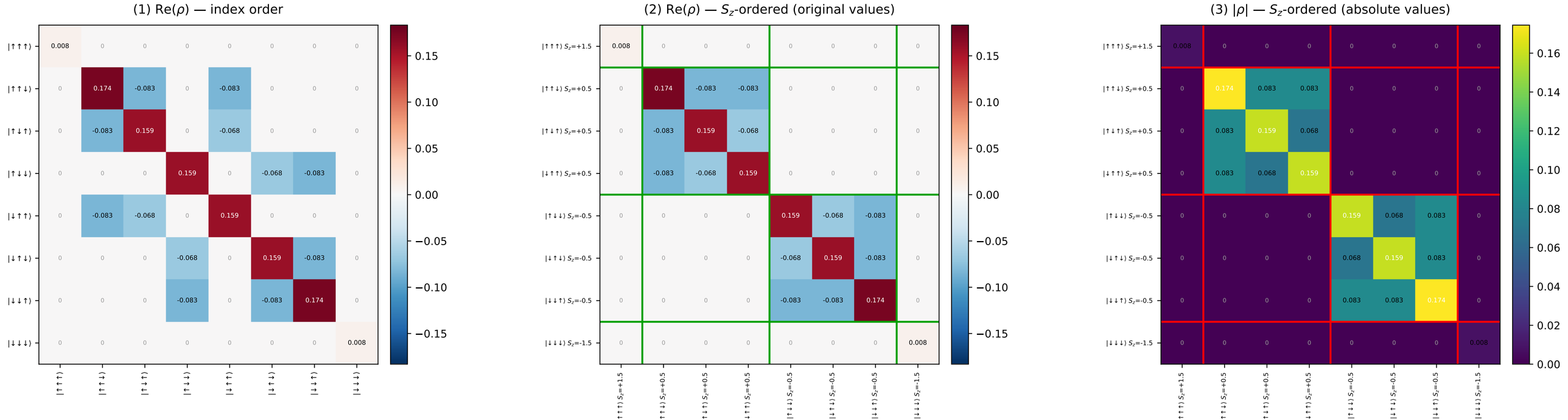
3-Body RDM (Sites 0,1,4) — same matrix in all three views



Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

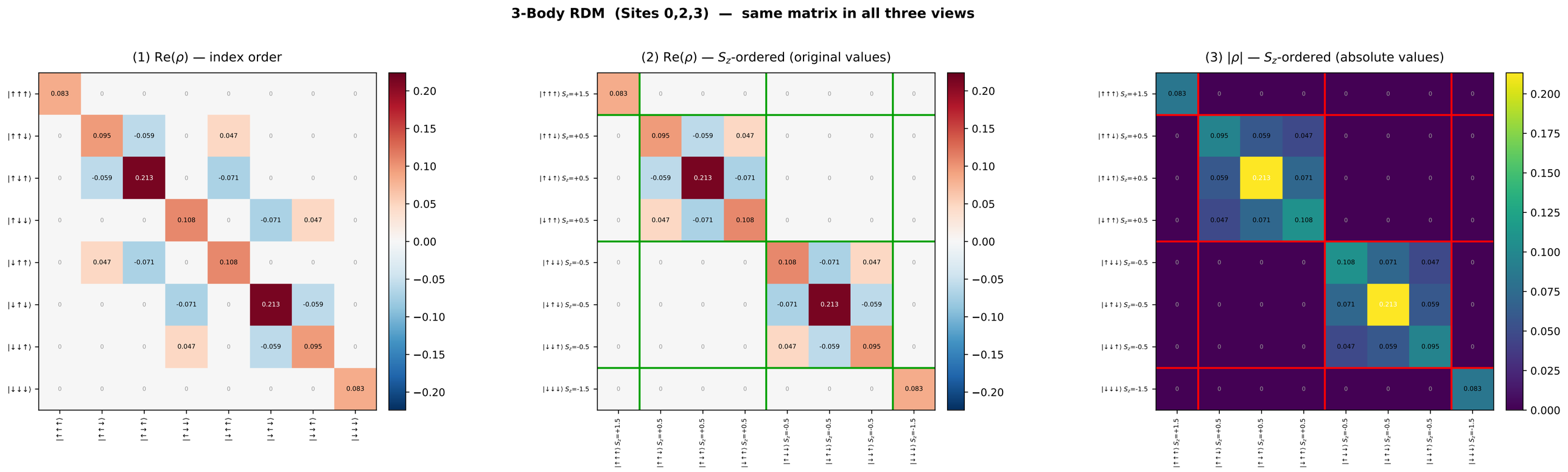
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000023 ≤ C_{UB} = 0.551931 verdict: GME

3-Body RDM (Sites 0,1,5) — same matrix in all three views

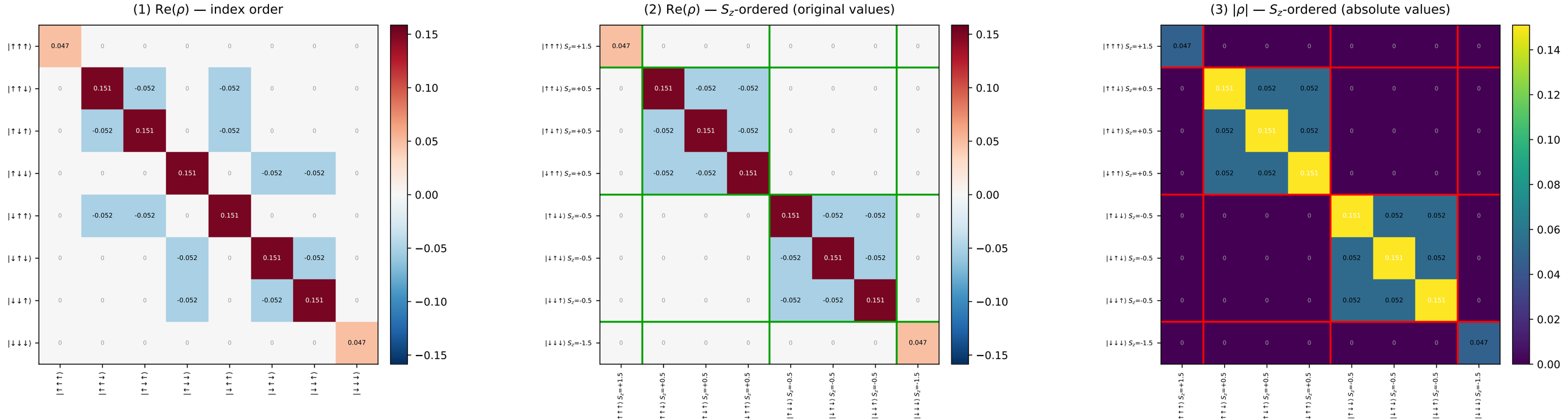


Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000022 ≤ C_{UB} = 0.399163 verdict: GME

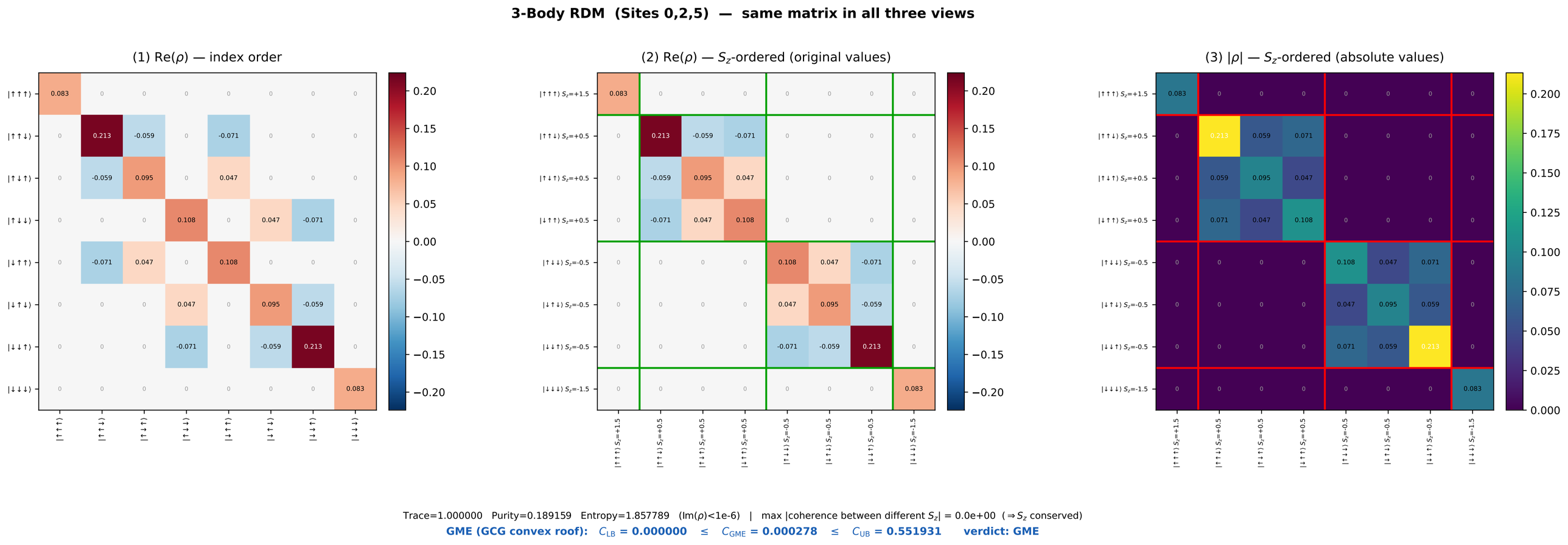

3-Body RDM (Sites 0,2,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000012 ≤ C_{UB} = 0.550327 verdict: GME

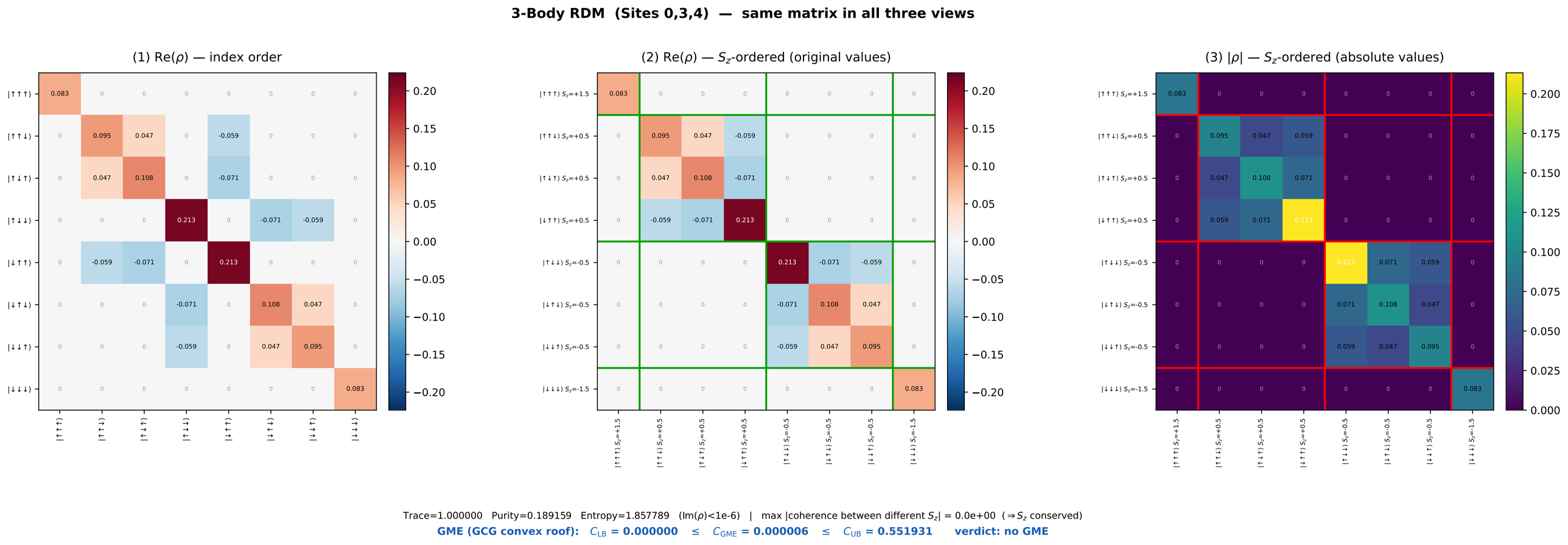
3-Body RDM (Sites 0,2,4) — same matrix in all three views

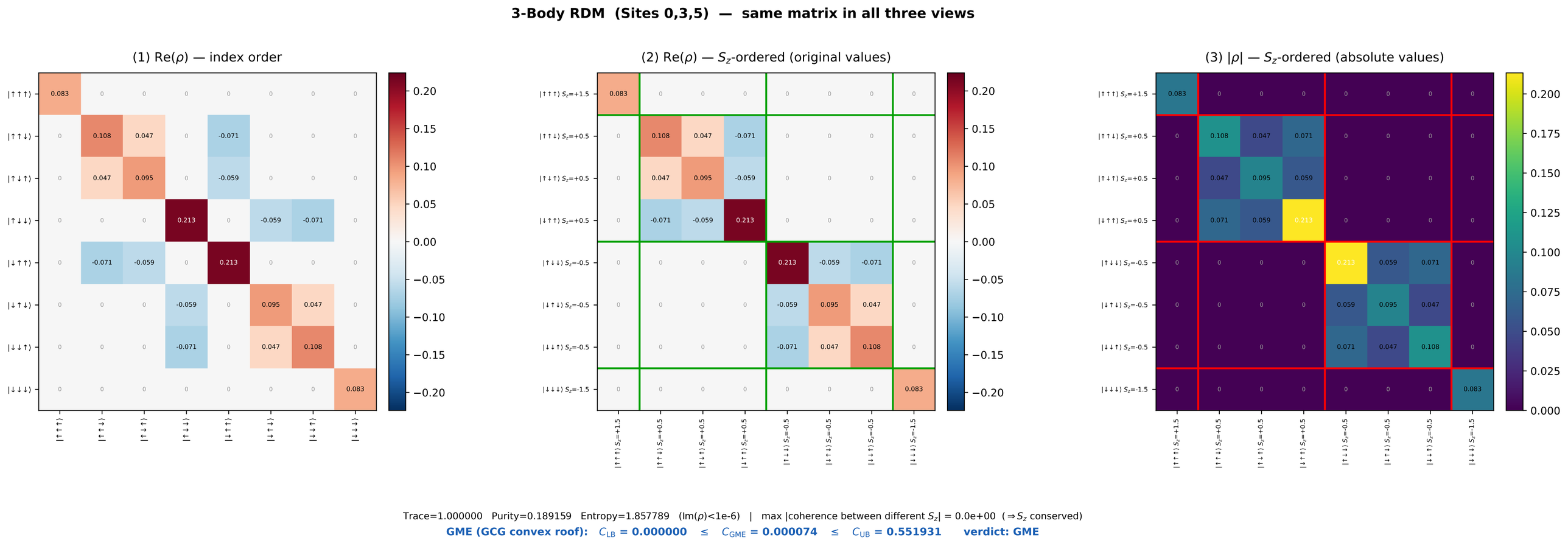


Trace=1.000000 Purity=0.173594 Entropy=1.869971 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

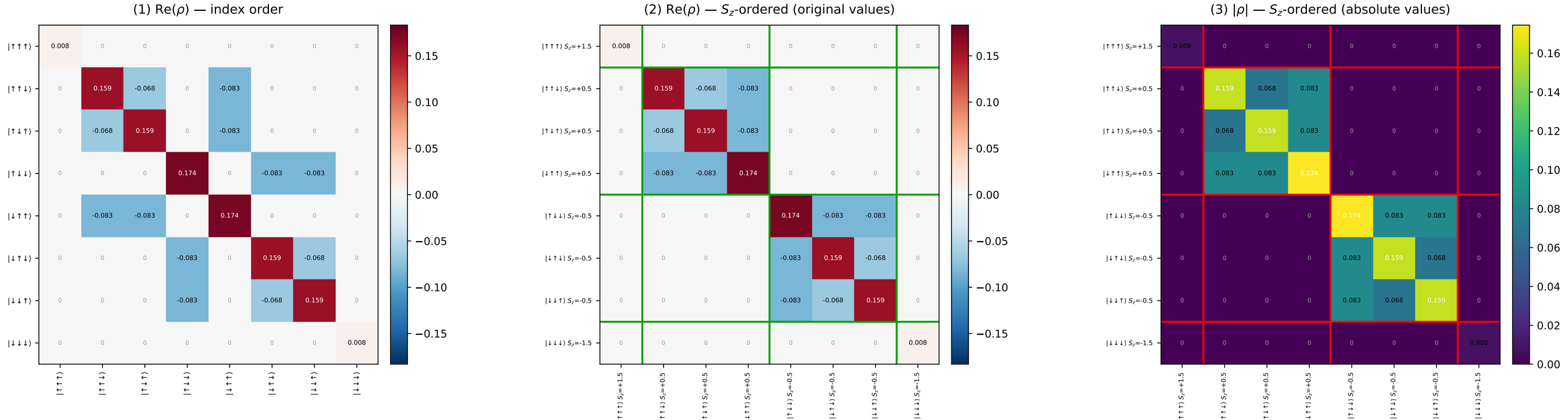
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000256 ≤ C_{UB} = 0.391258 verdict: GME


3-Body RDM (Sites 0,2,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000278 ≤ C_{UB} = 0.551931 verdict: GME


3-Body RDM (Sites 0,3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000006 ≤ C_{UB} = 0.551931 verdict: no GME


3-Body RDM (Sites 0,3,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000074 ≤ C_{UB} = 0.551931 verdict: GME

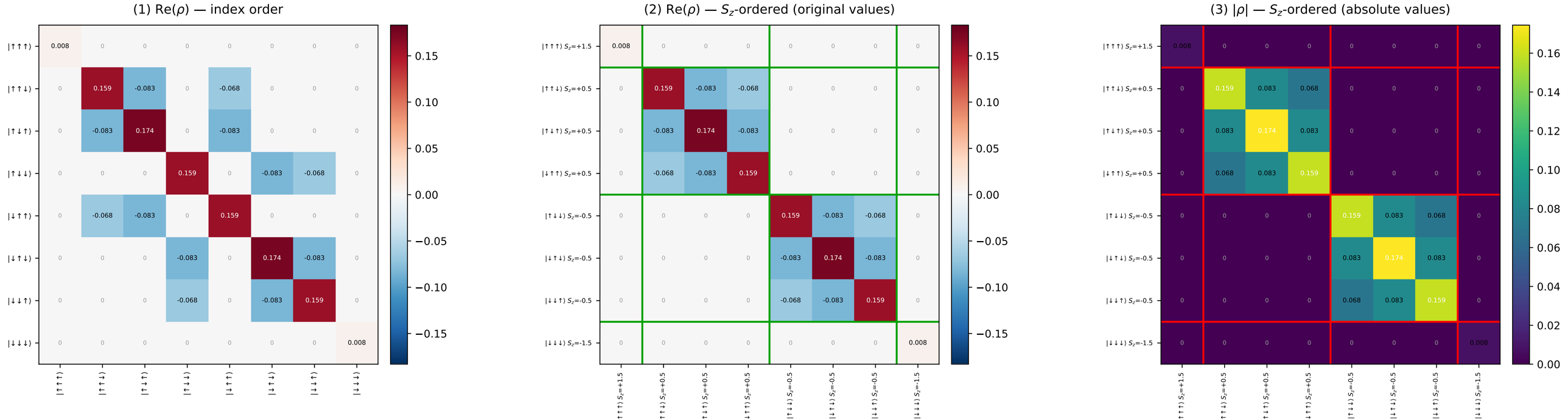
3-Body RDM (Sites 0,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000553 ≤ C_{UB} = 0.399153 verdict: GME

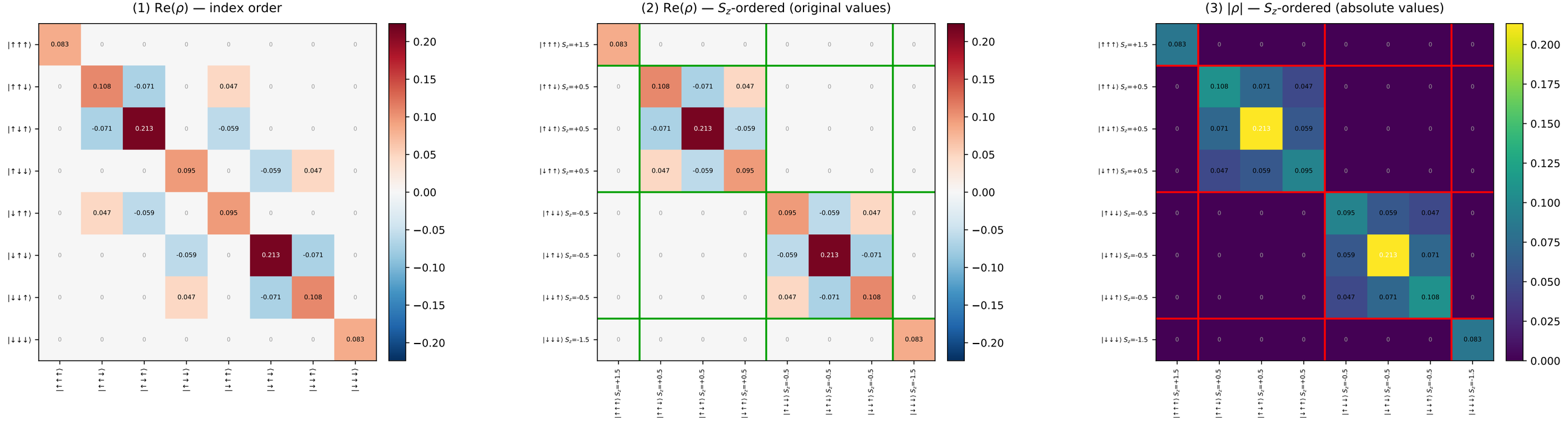
3-Body RDM (Sites 1,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001594 ≤ C_{UB} = 0.399163 verdict: GME

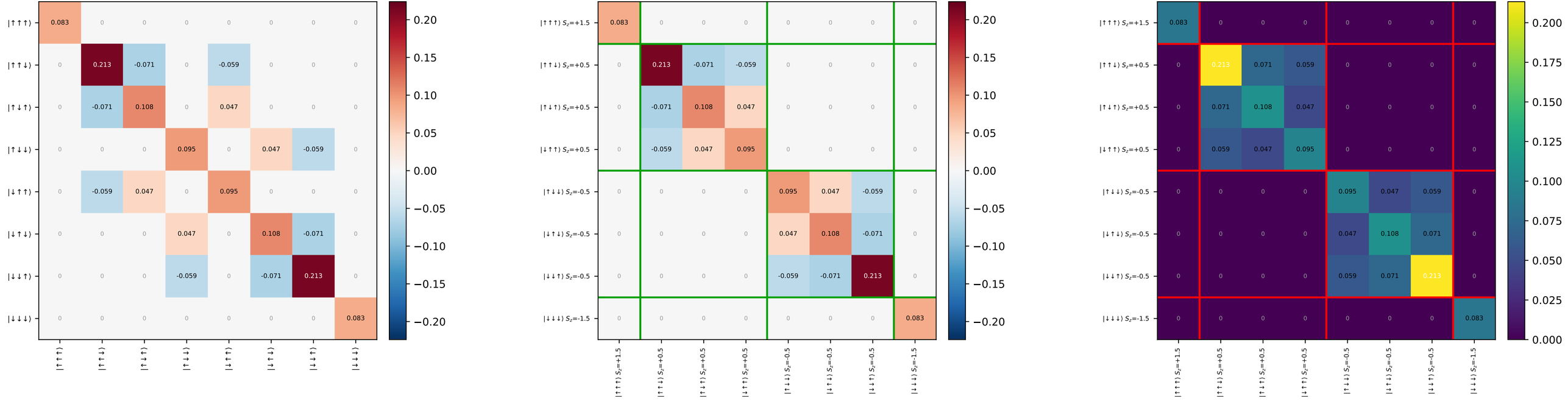
3-Body RDM (Sites 1,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000000 ≤ C_{UB} = 0.550740 verdict: no GME

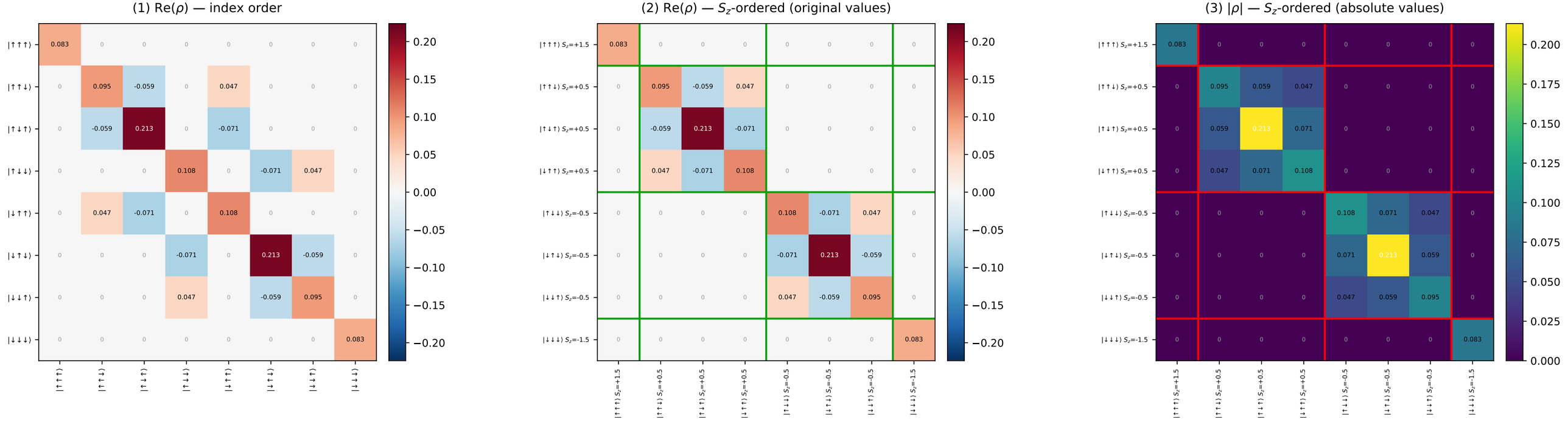
3-Body RDM (Sites 1,2,5) — same matrix in all three views



Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000007 ≤ C_{UB} = 0.551931 verdict: no GME

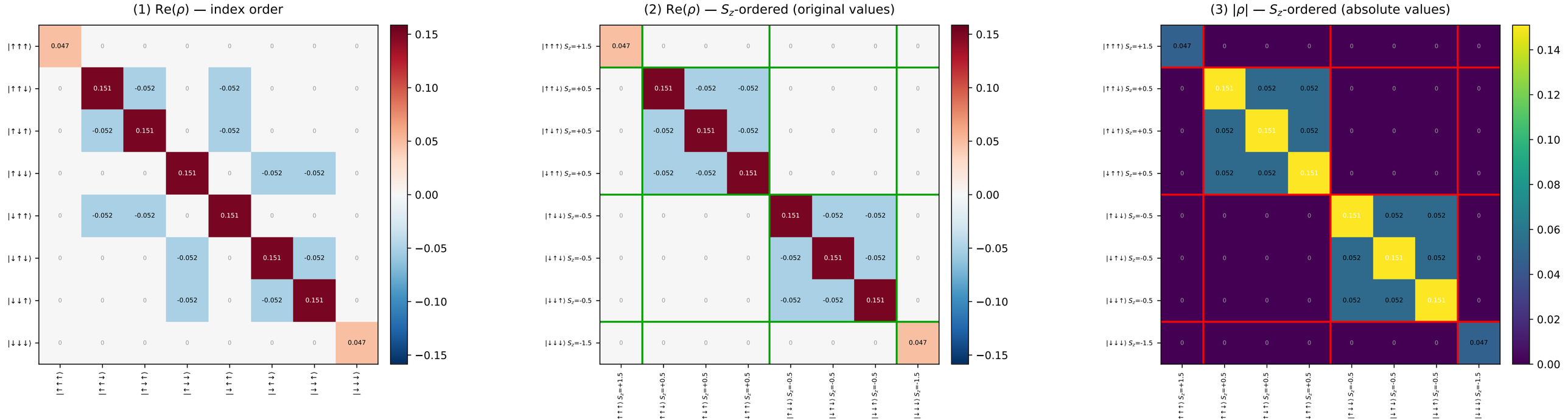
3-Body RDM (Sites 1,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000002 ≤ C_{UB} = 0.550809 verdict: no GME

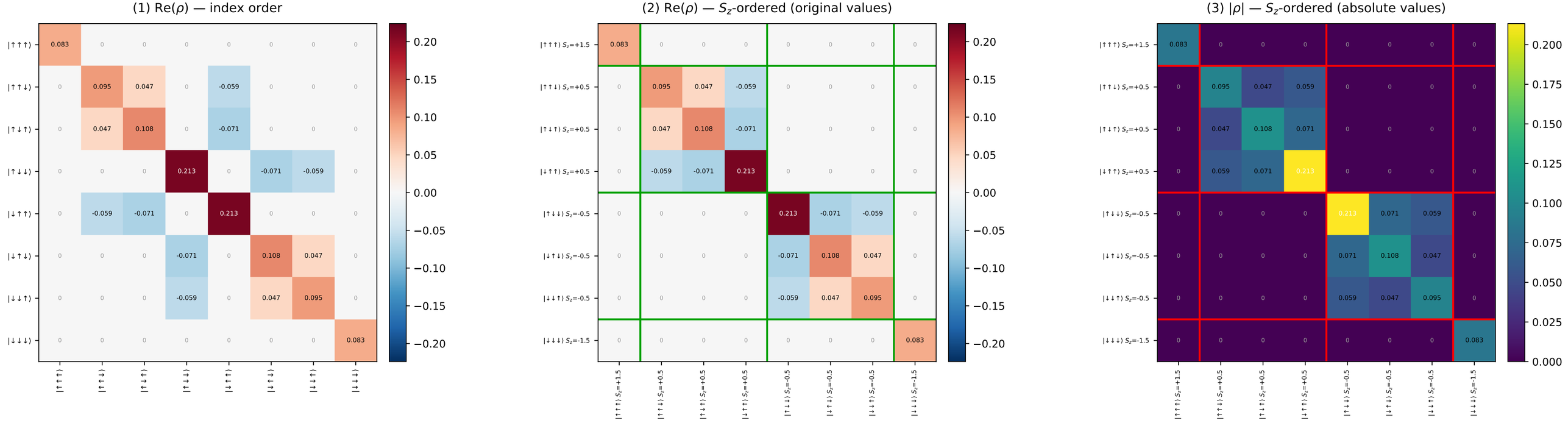
3-Body RDM (Sites 1,3,5) — same matrix in all three views



Trace=1.000000 Purity=0.173594 Entropy=1.869971 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000256 ≤ C_{UB} = 0.391258 verdict: GME

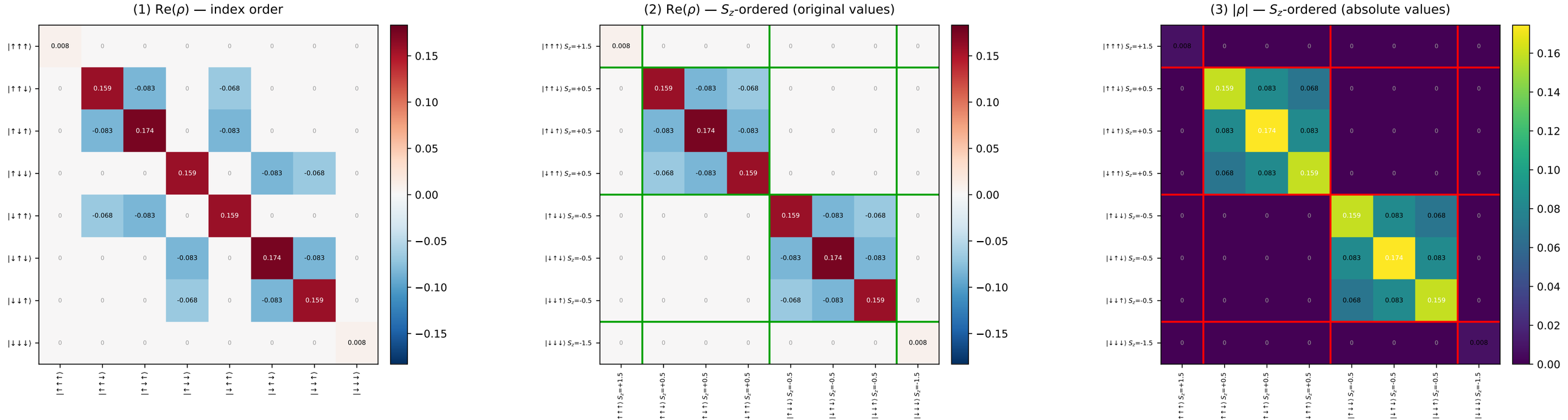
3-Body RDM (Sites 1,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000006 ≤ C_{UB} = 0.551931 verdict: no GME

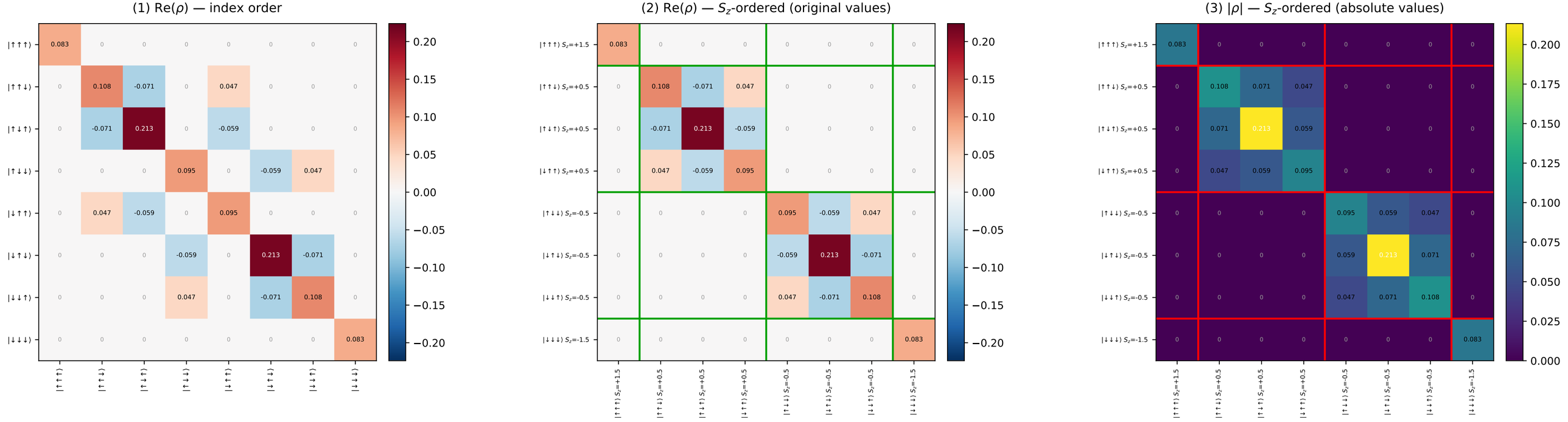
3-Body RDM (Sites 2,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

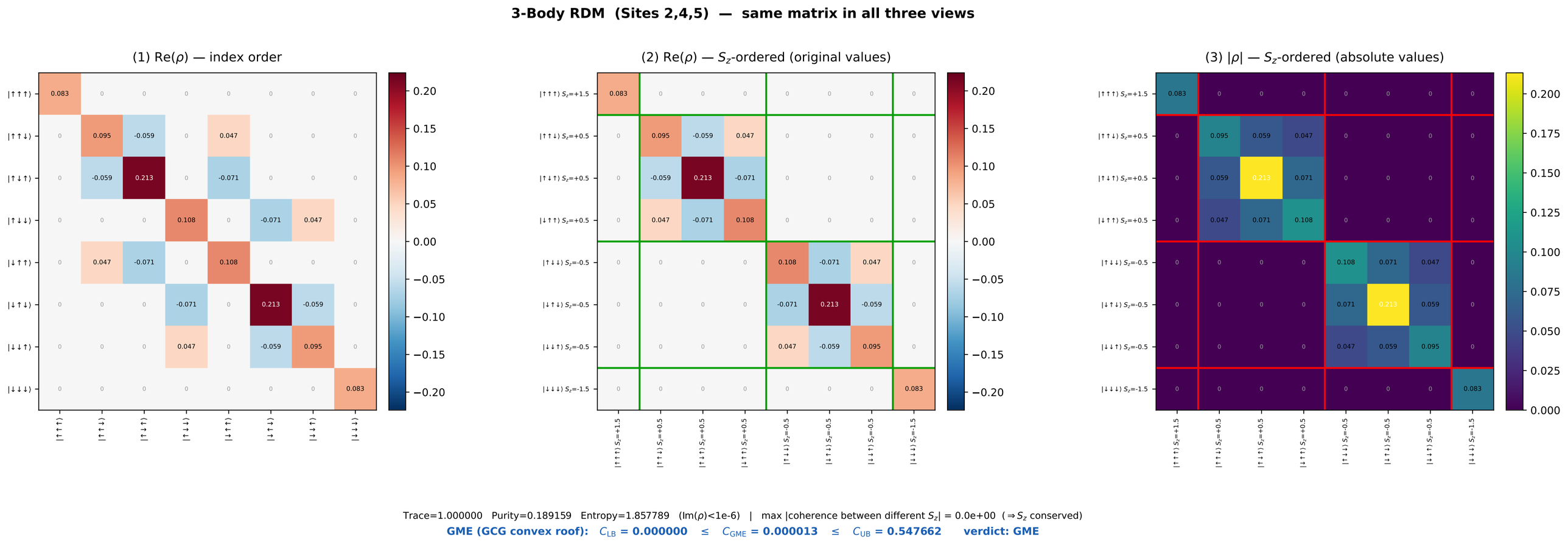
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001176 ≤ C_{UB} = 0.399160 verdict: GME

3-Body RDM (Sites 2,3,5) — same matrix in all three views

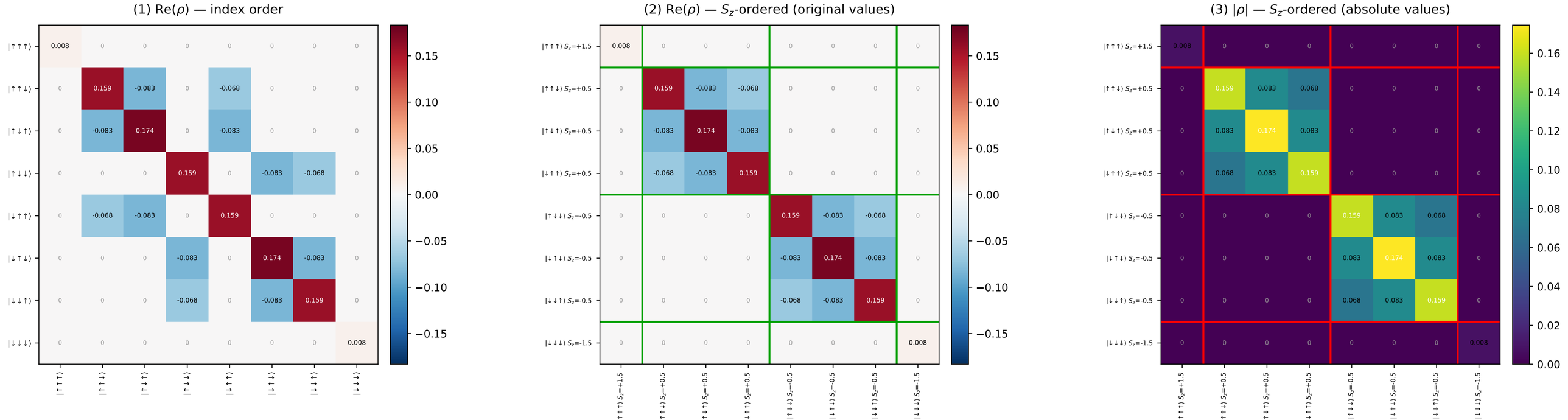


Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000008 ≤ C_{UB} = 0.551093 verdict: no GME


3-Body RDM (Sites 2,4,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.189159 Entropy=1.857789 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000013 ≤ C_{UB} = 0.547662 verdict: GME

3-Body RDM (Sites 3,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.235436 Entropy=1.526298 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000156 ≤ C_{UB} = 0.399059 verdict: GME